GRIFFIN KUCHAR

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EDUCATION

Texas Christian University — Fort Worth, TX
BS Computer Science — May 2027 Graduation
Minors: Mathematics, Economics — GPA: 3.8/4.0

EXPERIENCE

AMD — May 2025 – August 2025
System Validation Engineering Intern — *Austin, TX*

- Developed an LLM-powered full-stack **agentic** web application that enabled **20+** engineers across different teams to efficiently query and create Jira tickets, accelerating AMD-wide debugging workflows by **~50%**.
- Designed and executed stress-test and benchmarking workloads with **15+** unique components across **20+** systems simultaneously, validating AMD CPUs and resulting in a **~75%** increase in post-launch product success.

Texas Christian University & Children's Health Dallas — January 2026 – Present
AI/ML Researcher — *Fort Worth & Dallas, TX*

- Reduced average ICU transfer delay for at-risk pediatric patients by **23%** (from 8.4 to 6.4 hours) by training machine learning models on de-identified pediatric EHR data to detect decompensation **~87** minutes earlier than traditional threshold-based protocols.
- Engineered **516** features from 15,166 patient records by integrating time-series vitals, lab values, PEWS scores, clinical notes, and sepsis indicators across five data sources, applying SMOTE and calibration to address a 0.6% class imbalance across 90 decompensation cases.

Texas Christian University — August 2025 – Present
Computer Science Tutor — *Fort Worth, TX*

- Tutored **40+** first- and second-year computer science students across **three** programming courses, boosting average exam performance by **~10** points, and improving assignment completion rates.

PROJECTS

Agentic Ticketing Chatbot | *Flask (Python), ReactJS, PostgreSQL, OpenAI API* **| AMD**

- Designed and implemented a full-stack, **multi-agent AI system** enabling natural-language Jira ticket creation, querying, and workflow automation.
- Deployed and operated a cloud-hosted service used by **20+** AMD engineers, reducing debugging and ticketing turnaround time by **~50%**.
- Collaborated with a cross-functional team of **5** engineers, utilizing **Git/GitHub** for version control.
- Authored and maintained **5** technical documents, accelerating project onboarding and adoption.

Computer Vision Emotion Facial Detector| *Python, CUDA* **| TCU**

- Built a real-time emotion classifier utilizing CuPy CUDA kernels for GPU-accelerated computer vision image processing and a custom **2-layer CNN** trained on 20k+ images to classify 3 emotions from a live webcam video feed, recording a **3x** speedup in image preprocessing over its CPU sequential variant.

RELEVANT COURSES

Machine Learning, Data Structures and Algorithms, Web Technologies, OOP and Design Patterns, Database Systems, Operating Systems, GPU and Parallel Computing, Computer System Fundamentals, Computer Organization/Architecture, Discrete Mathematics II, Linear Algebra, Calculus II, Micro/Macroeconomics

HONORS & AWARDS

Dean's Honor List (5x) **|** Upsilon Pi Epsilon (UPE) TCU Chapter Inductee (Top 7 GPA) & Chapter Vice President **|** TCU Scholar (4.0 Semester) **|** Dean's Scholar **|** TCU Computer Science Society Featured Speaker

SKILLS & TOOLS

Languages: Java, Python, C, C++, JavaScript, TypeScript, SQL, HTML, CSS, Bash, Batch, Assembly (x86 & ARM)
Frameworks/Tools: Pandas, Numpy, Jupyter, React.js, Vue.js, Spring Boot, Flask, Django, MySQL, Git, Linux, AWS, Azure, Jira, Confluence, Docker, VSCode, CUDA, Postman, IntelliJ, PyCharm, GitHub, Cursor, Node.js, Box, Slack